NETWORK MANAGEMENT AND SECURITY

Computer networks are essential in modern organizations because they support communication, resource sharing, online transactions, cloud computing, and information exchange. As networks continue to grow in size and complexity, organizations require effective network management and strong security mechanisms to ensure reliability, efficiency, and protection of information resources.

Network management involves administering, monitoring, maintaining, and controlling network resources and devices. Network security focuses on protecting networks, systems, and data from unauthorized access, attacks, and damage.

These lecture notes discuss the following areas:

1. Network Management Issues and Challenges
2. User Profiling and Access Control
3. Network Security Concepts
4. Firewalls and Failure Recovery

NETWORK MANAGEMENT ISSUES AND CHALLENGES

Network management refers to the process of monitoring, administering, operating, and maintaining computer networks to ensure smooth and secure operation. Network administrators use management tools and policies to supervise devices such as routers, switches, servers, wireless access points, and firewalls.

The main goals of network management are:

- Ensuring network availability
- Improving network performance
- Enhancing security
- Detecting and correcting faults
- Optimizing resource utilization
- Supporting organizational communication

Functions of Network Management

Network management commonly follows the FCAPS model.

a) Fault Management

Fault management involves identifying, isolating, correcting, and recording network problems.

Examples of faults include:

- Server failure
- Router malfunction
- Broken network cables
- Software crashes
- Power outages

Importance:

- Minimizes downtime
- Improves reliability
- Maintains service continuity

b) Configuration Management

Configuration management focuses on maintaining proper device and system settings.

Activities include:

- Installing hardware and software
- Assigning IP addresses
- Updating configurations
- Backing up device settings

- Maintaining network documentation

Benefits:

- Simplifies troubleshooting
- Prevents configuration errors
- Enhances consistency

c) Accounting Management

Accounting management monitors and records network resource usage.

Examples:

- Monitoring internet bandwidth usage
- Tracking user activities
- Allocating network resources
- Supporting billing systems

Importance:

- Prevents misuse
- Supports budgeting
- Improves resource planning

d) Performance Management

Performance management ensures the network operates efficiently.

Performance indicators include:

- Bandwidth utilization
- Throughput
- Response time

- Packet loss
- Network latency

Benefits:

- Improves user experience
- Reduces congestion
- Enhances productivity

e) Security Management

Security management protects networks and systems from threats.

Activities include:

- Authentication
- Access control
- Encryption
- Monitoring attacks
- Security auditing

Importance:

- Protects sensitive information
- Prevents unauthorized access
- Reduces cyber risks

Network Management Issues and Challenges

a) Network Complexity

Modern networks consist of many interconnected devices, cloud systems, wireless technologies, and remote users. Managing these infrastructures is difficult because of:

- Multiple technologies

- Different vendors
- Large numbers of devices
- Integration problem

b) Scalability Problems

As organizations grow, networks expand in terms of users, devices, and services.

Challenges include:

- Increased traffic
- Performance degradation
- Difficult resource management
- Address management issues

c) Network Congestion

Congestion occurs when traffic exceeds available bandwidth.

Effects include:

- Slow internet speeds
- Delayed communication
- Packet loss
- Poor application performance

Causes:

- Excessive downloads
- Video streaming
- Malware traffic
- Many connected users

d) Security Threats

Networks face various cybersecurity threats such as:

- Viruses
- Worms
- Ransomware
- Phishing attacks
- Denial-of-service attacks
- Insider threats

Consequences include:

- Data loss
- Financial damage
- Service interruption
- Reputational damage

e) Downtime and Reliability Issues

Network downtime affects organizational operations.

Causes include:

- Hardware failure
- Human error
- Software bugs
- Cyberattacks
- Power failure

Impacts:

- Reduced productivity
- Loss of revenue
- Customer dissatisfaction

f) Wireless Network Challenges

Wireless networks introduce issues such as:

- Signal interference
- Unauthorized access
- Weak coverage
- Device mobility management

Strategies for Effective Network Management

Organizations can improve network management by:

- Continuous monitoring
- Regular maintenance
- Network documentation
- Employee training
- Using automation tools
- Capacity planning
- Implementing security policies

USER PROFILING AND ACCESS CONTROL

User Profiling

User profiling refers to the process of identifying users and maintaining information about their roles, responsibilities, access privileges, and behavior.

A user profile may contain:

- Username
- Password
- Department
- User role
- Access rights
- Login history
- Device information

Importance of User Profiling

a) Improves Security

Administrators can detect suspicious activities and unauthorized behavior.

b) Supports Access Management

Users are assigned permissions according to their responsibilities.

c) Enhances Accountability

Activities can be traced to individual users.

d) Improves User Experience

Systems can provide customized services and settings.

e) Assists Auditing

User records support investigations and compliance requirements.

Access Control

Access control refers to mechanisms used to regulate who can access systems, applications, data, and network resources.

The objective is to allow authorized access while preventing unauthorized access.

Elements of Access Control

a) Identification

Users identify themselves using usernames, IDs, or biometrics.

b) Authentication

Authentication verifies the identity of users.

Methods include:

- Passwords
- PINs
- Smart cards
- Fingerprints
- Facial recognition
- One-time passwords (OTP)

c) Authorization-Authorization determines what resources a user can access.

d) Accountability-User activities are logged and monitored.

Access Control Models

a) Discretionary Access Control (DAC)-The owner of a resource decides who can access it.

b) Mandatory Access Control (MAC)-Access is controlled using security classifications.

c) Role-Based Access Control (RBAC)-Permissions are assigned according to user roles.

Access Control Techniques

a) Password Policies

Organizations enforce:

- Strong passwords
- Password expiry
- Minimum password length
- Password complexity rules

b) Multi-Factor Authentication (MFA)-Users provide more than one authentication factor.

c) Biometric Authentication-Uses physical or behavioral characteristics.

NETWORK SECURITY

Network security refers to policies, technologies, and practices designed to protect network systems, devices, and information from attacks, unauthorized access, and damage.

Objectives of Network Security

The major objectives are represented by the CIA Triad.

a) Confidentiality

Ensures information is accessible only to authorized individuals.

Methods used:

- Encryption
- Passwords
- Access controls

Example:

Protecting examination results from unauthorized users.

b) Integrity

Ensures information remains accurate and unaltered.

Methods:

- Hashing
- Checksums
- Digital signatures

Example:

Preventing unauthorized modification of financial records.

c) Availability

Ensures systems and services remain accessible whenever required.

Methods:

- Data backups
- Redundancy
- Disaster recovery
- Load balancing

Example:

Keeping e-learning systems operational.

Common Network Threats

a) Malware

Malware refers to malicious software designed to damage or disrupt systems.

Types include:

- Viruses
- Worms
- Trojan horses
- Spyware

- Ransomware

b) Phishing Attacks

Attackers trick users into revealing confidential information using fake emails, websites, or messages.

c) Denial of Service (DoS) Attacks

Attackers overload systems with excessive traffic to make services unavailable.

Effects:

- Downtime
- Slow performance
- Business disruption

d) Man-in-the-Middle (MITM) Attacks

Attackers secretly intercept communication between users.

Risks:

- Data theft
- Credential interception
- Session hijacking

e) Insider Threats

Trusted individuals misuse access privileges.

Examples:

- Data theft
- Unauthorized disclosure
- Sabotage

Security Mechanisms

a) Encryption

Encryption converts readable data into unreadable form.

Benefits:

- Protects confidentiality
- Secures communication

b) Antivirus Software

Detects and removes malware.

Functions:

- Real-time scanning
- Malware detection
- File quarantine

c) Intrusion Detection Systems (IDS)

Monitor networks for suspicious activities.

Types:

- Network-based IDS
- Host-based IDS

d) Virtual Private Networks (VPNs)

VPNs create secure encrypted communication channels for remote users.

Benefits:

- Secure remote access
- Protection on public networks